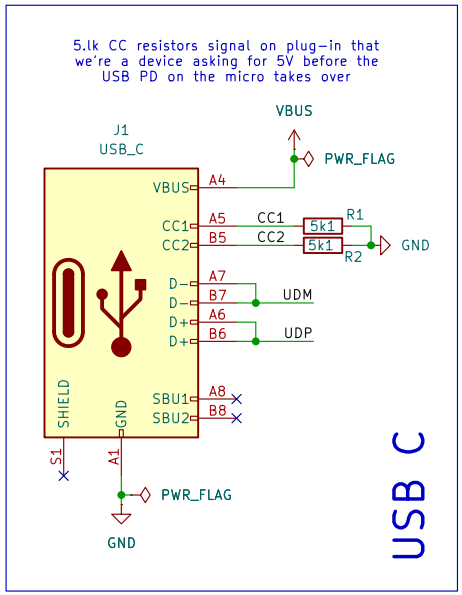
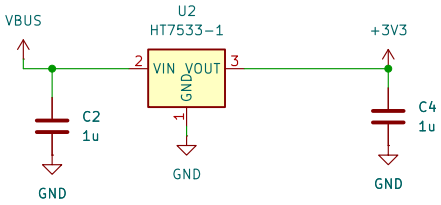
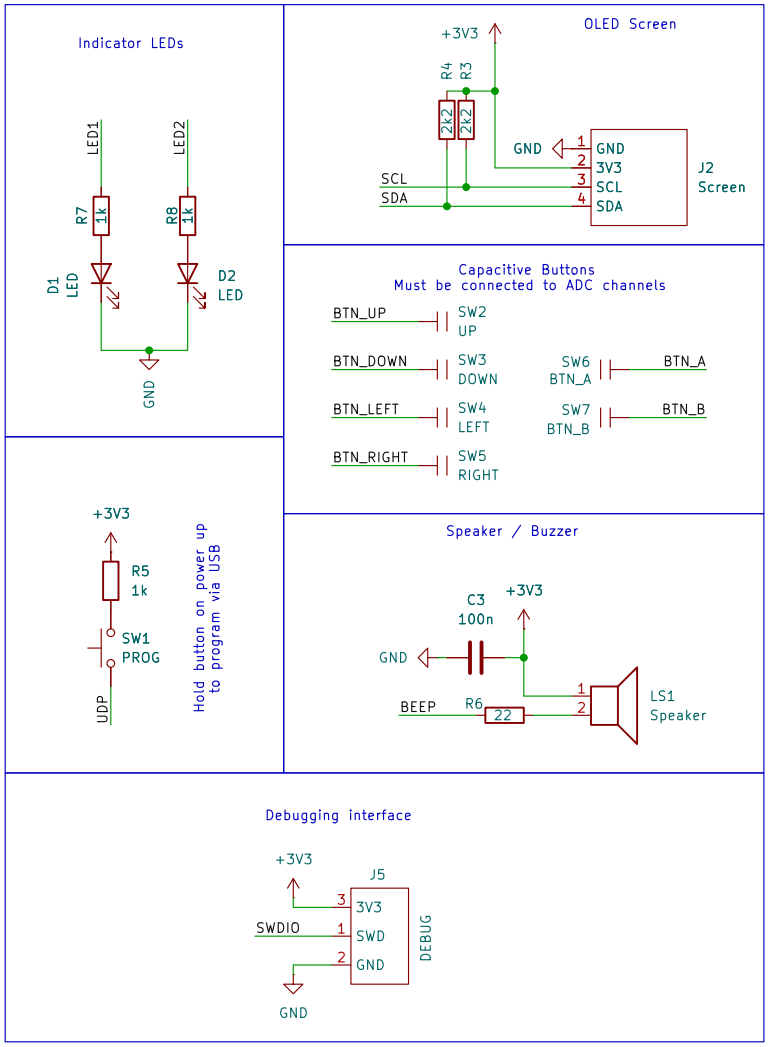
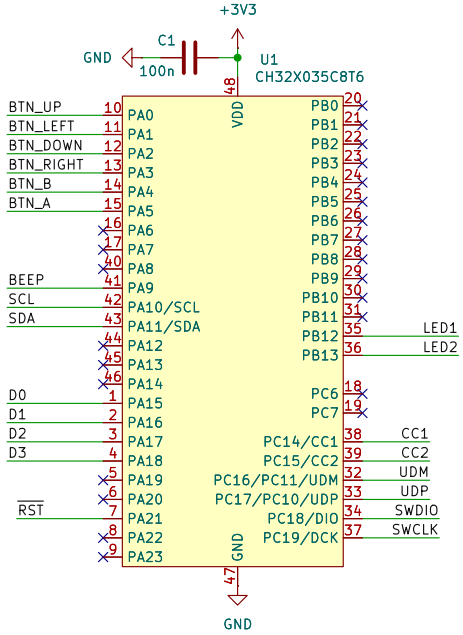




Note that VBUS could get up to or exceed 20V if the micro makes appropriate request via USB PD. This regulator might struggle at this point, especially when driving the buzzer





Possible firmware:
- Game console
- Timer
- Game pad via USBHID
- Mouse wiggler
- IR / HID bridge if we have IR LED/PD
- Ticker display via USBCDC
- USB PD Power supply
- I/O tool with exposed pads
- Something using PGA/OPA!
- Some resistors for a transistor tester?

Matthew Chilcott
University of Otago

Sheet: /
File: 2025ClassProject.kicad_sch

Title: PHSI245 Class Project

Size: A4	Date: 2025-07-22	**Rev:**
KiCad E.D.A. 10.0.3		Id: 1/1